

Lec 3-2, Sep 17 (Wed), 2025

- Relative rates of growth & Big-oh and little-oh notations
- Related rates (3.8)
- extreme values of functions (4.1)
- find local/global extrema

Relative rates of growth

Def: Let f & g be functions which are positive for all sufficiently large inputs. We say that as $x \rightarrow \infty$, f grows

(i) faster than g ;

(ii) slower than g ;

(iii) at the same rate as g ;

if $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} :$

(i) $= \infty$;

(ii) $= 0$;

(iii) $= L \in (0, \infty)$, (i.e., $L \in \mathbb{R}_{>0}$).

Rmk: Usually, f and g above are nondecreasing, but they do not have to be.

e.g. (a) If f is eventually positive and $k \in \overset{(0, \infty)}{=} \mathbb{R}_{>0}$ is fixed,

$$\text{then } \lim_{x \rightarrow \infty} \frac{k \cdot f(x)}{f(x)} = \lim_{x \rightarrow \infty} k = k \in (0, \infty),$$

so $k \cdot f$ and f grow at the same rate.

← constant function

(b) x^n vs K : For any fixed $n > 0$,

$$\lim_{x \rightarrow \infty} \frac{x^n}{K} = \infty$$

So x^n grows faster than K .

(b) x^m vs x^n (where $m > n > 0$)

$$\lim_{x \rightarrow \infty} \frac{x^m}{x^n} = \lim_{x \rightarrow \infty} x^{m-n} = \infty, \text{ so } x^m \text{ grows faster than } x^n.$$

• Growing at the same rate is a transitive relation!

If f and g grow at the same rate, and g and h also grow at the same rate, then so do f and h , since

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = L_1 > 0 \quad \text{and} \quad \lim_{x \rightarrow \infty} \frac{g(x)}{h(x)} = L_2 > 0$$

(where L_1, L_2 are finite)

$$\Rightarrow \lim_{x \rightarrow \infty} \frac{f(x)}{h(x)} = \lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} \cdot \frac{g(x)}{h(x)} = \lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} \lim_{x \rightarrow \infty} \frac{g(x)}{h(x)} = L_1 L_2 > 0.$$

$\Rightarrow f$ and h grow at the same rate!

□

e.g. $\sqrt{x^2+2025}$ and $(98\sqrt{x}-1)^2$ grow at the same rate ,

Since they both grow at the same rate as x (when x goes infinity):

$$\lim_{x \rightarrow \infty} \frac{\sqrt{x^2+2025}}{x} = \lim_{x \rightarrow \infty} \sqrt{\frac{x^2+2025}{x^2}} = \lim_{x \rightarrow \infty} \sqrt{1+\frac{2025}{x^2}} = 1 ,$$

$$\lim_{x \rightarrow \infty} \frac{(98\sqrt{x}-1)^2}{x} = \lim_{x \rightarrow \infty} \frac{98^2 x - 2 \cdot 98\sqrt{x} + 1}{x} = 98^2 .$$

Similar ideas can be used to talk about the rates at which the functions approach 0 :

$$\text{e.g.: } \bullet \lim_{x \rightarrow 0} \sin x = \lim_{x \rightarrow 0} x = 0 , \quad \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \in (0, \infty)$$

$\Rightarrow f(x) = \sin x$ and $g(x) = x$ approaches 0 at the same rate as $x \rightarrow 0$.

$$\bullet \lim_{x \rightarrow 2} \frac{(x-2)^2}{x-2} = \lim_{x \rightarrow 2} (x-2) = 0$$

$\Rightarrow f(x) = (x-2)^2$ approaches 0 faster than $g(x) = x-2$, as $x \rightarrow 2$.

• Big-Oh and Little-Oh [Chap. 7.8]

Def : Let f and g be functions, both positive for sufficiently large x . Then we write :

• $f(x) = o(g(x))$ as $x \rightarrow \infty$ if $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = 0$. (1)

• $f(x) = O(g(x))$ as $x \rightarrow \infty$ if $\exists N, M \in \mathbb{R}$, such that

$$\frac{f(x)}{g(x)} \leq M, \quad \forall x \in [N, \infty). \quad (2)$$

$$\updownarrow \\ f(x) \leq M \cdot g(x)$$

Rmks

• These are called little-Oh and big-Oh notation, respectively.

• (1) states " f grows slower than g as $x \rightarrow \infty$ ", while

(2) states " f grows at most as fast as g , as $x \rightarrow \infty$ ".

• If $f(x) = o(g(x))$ as $x \rightarrow \infty$, then $f(x) = O(g(x))$ as $x \rightarrow \infty$.

(Why?)

• If $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = k \in \mathbb{R}_{>0}$, then $f(x) = O(g(x))$ as $x \rightarrow \infty$.

(Why?)

Qualitative.

e.g. (a) $k = O(1)$.

(b) More generally,

$$k f(x) = O(f(x)), \text{ if } f \text{ is eventually positive.}$$

(c) $a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0 = O(x^n)$ if $a_n > 0$, since

$$\lim_{x \rightarrow \infty} \frac{\sum_{i=0}^n a_i x^i}{x^n} = \lim_{x \rightarrow \infty} \sum_{i=0}^n a_i x^{i-n} = \sum_{i=0}^n a_i \lim_{x \rightarrow \infty} x^{i-n} = a_n$$

showing that $\sum_{i=0}^n a_i x^i$ and x^n have the same growth rate.

(d) $x + \sin x = O(x)$, since

$$\lim_{x \rightarrow \infty} \frac{x + \sin x}{x} = \lim_{x \rightarrow \infty} \left(1 + \frac{\sin x}{x} \right) = 1,$$

showing that $x + \sin x$ and x have the same growth rate.

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0}{x^n} &= \lim_{x \rightarrow \infty} \left(a_n + a_{n-1} x^{-1} + a_{n-2} x^{-2} + \dots + a_1 x^{1-n} + a_0 x^{-n} \right) \\ &= \left(\lim_{x \rightarrow \infty} a_n \right) + a_{n-1} \cdot \lim_{x \rightarrow \infty} x^{-1} + \dots + \lim_{x \rightarrow \infty} a_0 x^{-n} \\ &= a_n + 0 + \dots + 0 \\ &= a_n. \end{aligned}$$

• Related rates

Eg. Water runs into a conical tank at the rate of $\frac{1}{4} \text{ m}^3/\text{min}$.

The tank stands point down and has a height of 3 m and a base radius of 1.5 m.

Q: How fast is the water level rising when the water is 1.8 m deep?

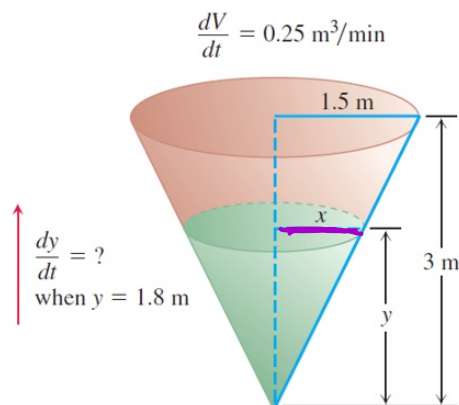


FIGURE 3.32 The geometry of the conical tank and the rate at which water fills the tank determine how fast the water level rises (Example 1).

Sol': Step 1. Name all the variables and assume they are differentiable functions of time t .

- time t
- Volume $V(t)$, height $y(t)$, radius of the water level $x(t)$.

Step 2. Write down the information and the quantity we want to compute.

- $\frac{dV}{dt} = \frac{1}{4} \text{ (m}^3/\text{min)}.$
- $\left. \frac{dy}{dt} \right|_{h=1.8 \text{ m}} = ?$
- $V(t) \xrightarrow[\text{formula}]{\text{volume}} \frac{1}{3} \pi (x(t))^2 y(t) = \frac{1}{3} \pi \left(\frac{y(t)}{2} \right)^2 y(t). \quad (*)$

Step 3. Relate these variables

$$(*) \Rightarrow \frac{1}{12} y(t)^3 \pi - V(t) = 0, \quad \forall t$$

Step 4. Take the derivative !

$$\frac{1}{12} \cdot 3 y(t)^2 \cdot y'(t) \pi - V'(t) = 0, \quad \forall t.$$

$$\Rightarrow y'(t) = \frac{V'(t)}{\frac{1}{4} \pi y(t)^2}, \quad \forall t.$$

Let $y(t) = 1.8 \text{ m}$. Then $y'(t) \Big|_{h=1.8 \text{ m}} = \frac{V'(t)}{\frac{1}{4} \pi y(t)^2} \Big|_{h=1.8 \text{ m}}$

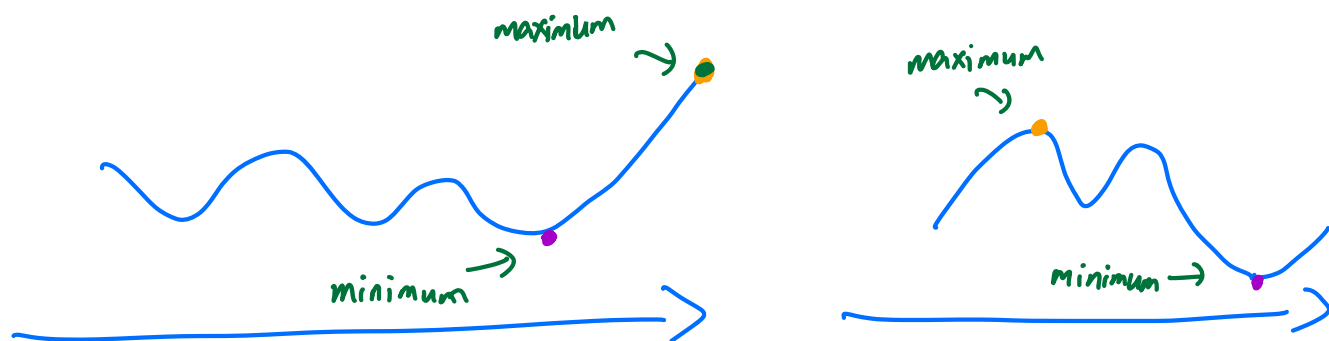
$$= \frac{\frac{1}{4}}{\frac{1}{4} \pi 1.8^2}$$

$$= \frac{25}{81 \pi} \text{ (m/min)}.$$

□.

Chap 4. Applications of Derivatives.

§4.1. Extreme values of functions



Definition: let f be a function with domain D . Then f has an

absolute maximum value on D at a point c if

$$f(c) \geq f(x), \forall x \in D.$$

and an absolute minimum value on D at c if

$$f(c) \leq f(x), \forall x \in D.$$

Rmks. ① Absolute maximum/minimum are also called global maximum/minimum.

② Extremum means maximum or minimum.

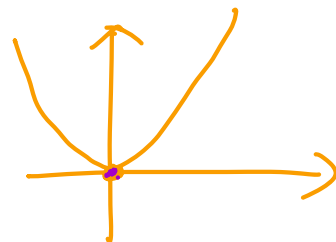
③ Plural forms: maxima, minima, extrema.

e.g. $f: D \rightarrow \mathbb{R}, f(x) = x^2$

• $D = \mathbb{R}$:

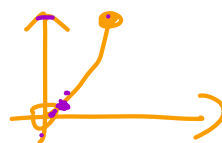
• No absolute maximum.

• absolute minimum at 0: value $f(0) = 0$.



• $D = [0, 3]$: absolute maximum at 3 : Value $f(3) = 9$

• No absolute minimum.



• $D = [0, 3]$: absolute maximum at 3 : Value $f(3) = 9$

• absolute minimum at 0 : Value $f(0) = 0$.



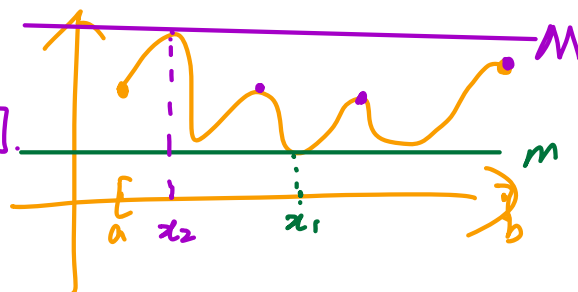
Thm 4.1.1 [Extreme Value Theorem].

If f is continuous on a closed interval $[a, b]$, then

f attains both an absolute maximum value M and an absolute minimum value m in $[a, b]$. That is, there exists numbers x_1 and x_2 in $[a, b]$, s.t. $f(x_1) = m$ and $f(x_2) = M$, and

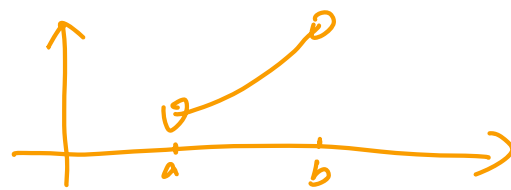
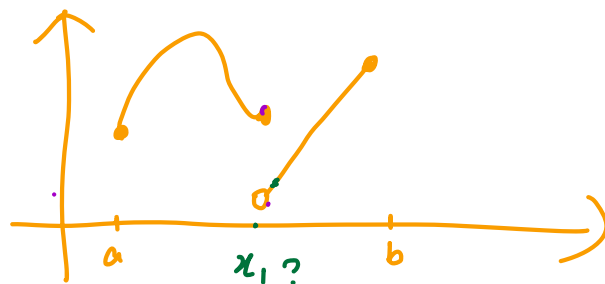
$$\underbrace{m}_{f(x_1)} \leq f(x) \leq \underbrace{M}_{f(x_2)} \text{ for any } x \in [a, b].$$

Pf: Omitted [requires some technical discussions].



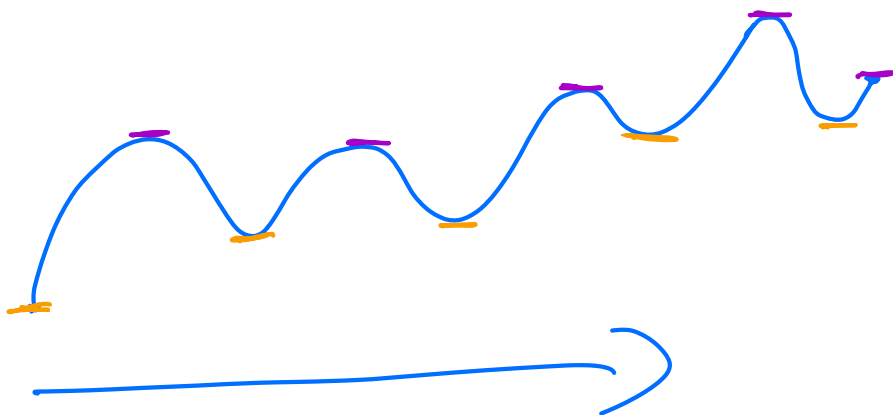
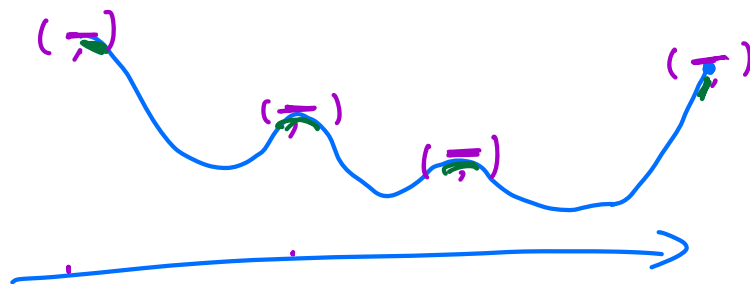
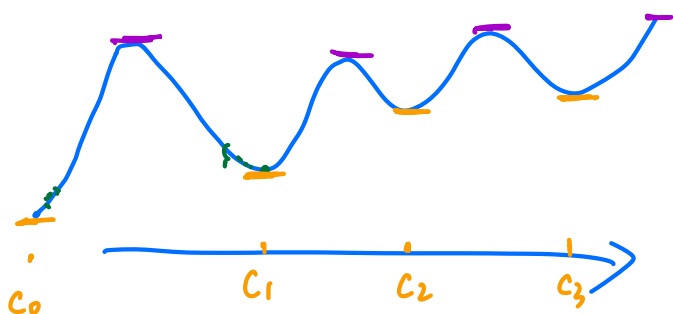
Rmk: f may not attain extrema if

- f is not continuous, or
- the interval is not closed.

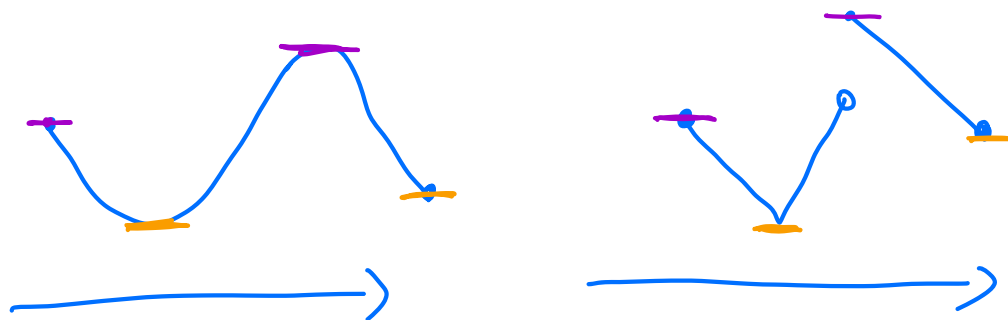


Definition : Let $f: D \rightarrow \mathbb{R}$.

- The function f is said to have a **local maximum** at c if there exists $\delta > 0$ s.t. $f(c) \geq f(x)$ for any $x \in (c-\delta, c+\delta) \cap D$.
- The function f is said to have a **local minimum** at c if there exists $\delta > 0$ s.t. $f(c) \leq f(x)$ for any $x \in (c-\delta, c+\delta) \cap D$.



Q: How to find local or absolute extrema?



Observe:

- extrema are **endpoint** ; or
- extrema have a **zero derivative** ; or
- extrema do not have **derivatives**.

Definition. Let $f: D \rightarrow \mathbb{R}$ and let c be an interior point of D .

Then c is a **critical point** of f if

- (i) $f'(c) = 0$, or ;
- (ii) $f'(c)$ does not exist finitely (i.e, as a real number in $\mathbb{R}$).

e.g.3. What are all the critical points of the function

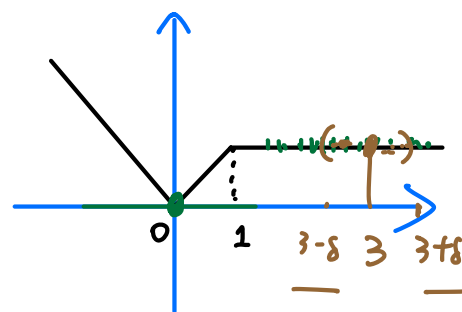
$$f(x) = \begin{cases} |x|, & \text{if } x < 1; \\ 1, & \text{if } x \geq 1. \end{cases}$$

what about local extrema?

Sol: (i) Solve for all the c , s.t. $f'(c) = 0$

$$\Rightarrow c \in (1, \infty).$$

(ii) Solve for all the c , s.t. $f'(c)$ does not exist finitely



$$\{0\} \cup \{1\} \cup (1, \infty)$$

$$\Rightarrow c \in \{0, 1\}$$

$\therefore$ all the critical points are given by $\{0\} \cup [1, \infty)$.

Additionally, from the graph, we know all the

local extrema are also given by $\{0\} \cup [1, \infty)$. $\square$.

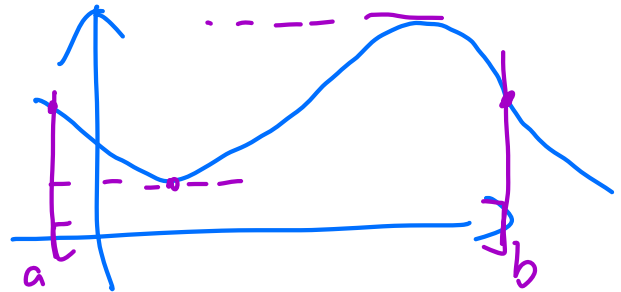
Thm 4.1.2 [First Derivative Theorem for Local Extreme Values].

If f has a local extremum value at an interior point c of its domain, and if f' is defined at c , then

$$f'(c) = 0.$$

The theorem above can as be rephrased as follows:

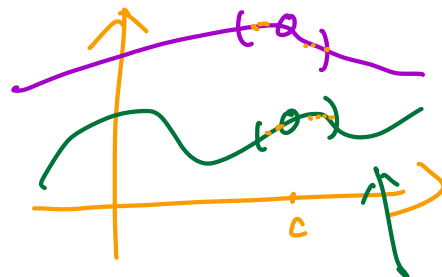
Let c be an interior point of D . If a function $f: D \rightarrow \mathbb{R}$ has a local extremum at c , then c is a critical point of f .



Before proving this, recall "limits preserve order".

Thm 2.2.5. If $f(x) \leq g(x)$, $\forall x$ in some open interval containing c , except possibly at $x=c$ itself, and the limits of f and g both exist as x approaches c , then

$$\lim_{x \rightarrow c} f(x) \leq \lim_{x \rightarrow c} g(x).$$

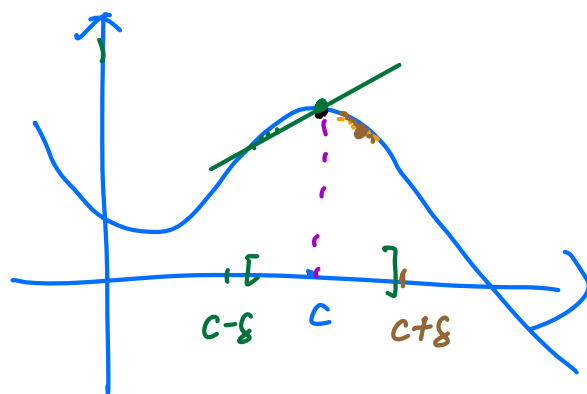


Pf of Thm 4.1.2. Suppose f has a local max at c ; the case of min is similar.

Step 1. write down the definition!

$\exists \delta > 0$, s.t

$$f(x) \leq f(c), \quad \forall x \in (c-\delta, c+\delta).$$



Step 2. compute the derivative from right!

For $x \in (c, c+\delta)$ we have

$$\left\{ \begin{array}{l} f(x) \leq f(c) \\ x - c > 0 \end{array} \right.$$

$$\therefore \frac{f(x) - f(c)}{x - c} \leq 0.$$

$$\therefore \lim_{x \rightarrow c^+} \frac{f(x) - f(c)}{x - c} \leq 0$$

by Thm 2.2.5.

Step 3. Compute the derivative from left !

For $x \in (c-\delta, c)$, we have

$$\begin{cases} f(x) \leq f(c) \\ x < c \end{cases}$$

$$\therefore \frac{f(x) - f(c)}{x - c} \geq 0.$$

$$\therefore \lim_{x \rightarrow c^-} \frac{f(x) - f(c)}{x - c} \geq 0$$

by Thm 2.2.5.

Step 4. combine !

$$0 \leq \lim_{x \rightarrow c^-} \frac{f(x) - f(c)}{x - c} = f'_-(c) = f'(c) = f'_+(c) = \lim_{x \rightarrow c^+} \frac{f(x) - f(c)}{x - c} \leq 0$$

$$\Rightarrow f'(c) = f'_-(c) = f'_+(c) = 0.$$

□

Def. The normal line to a curve at (x_0, y_0) is the line perpendicular

to the tangent line to the curve at (x_0, y_0) . [slope of the normal line
= $\frac{-1}{\text{slope of the tangent line}}$]

e.g. For $x^3 + y^3 - 6xy = 0$, at point $(3,3)$,

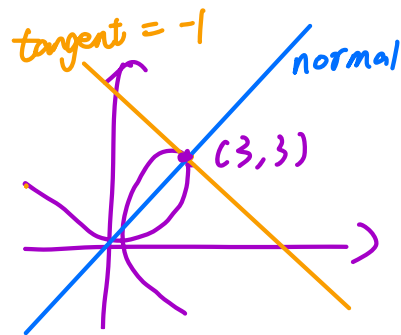
we already know the slope of the tangent line is -1 .

Hence, the slope of the normal line is $\frac{-1}{\text{slope of the tangent line}}$
 $= \frac{-1}{-1} = 1$

The normal line is

$$y - 3 = 1 \cdot (x - 3) = x.$$

Q.



Thm 4.1.2. interior + local extremum $\Rightarrow$ critical

or

interior + diff' + local extremum $\Rightarrow f'(c) = 0$.

Counterexample. The converse of the Theorem 4.1.2 is

not true; having a critical point at c does not imply

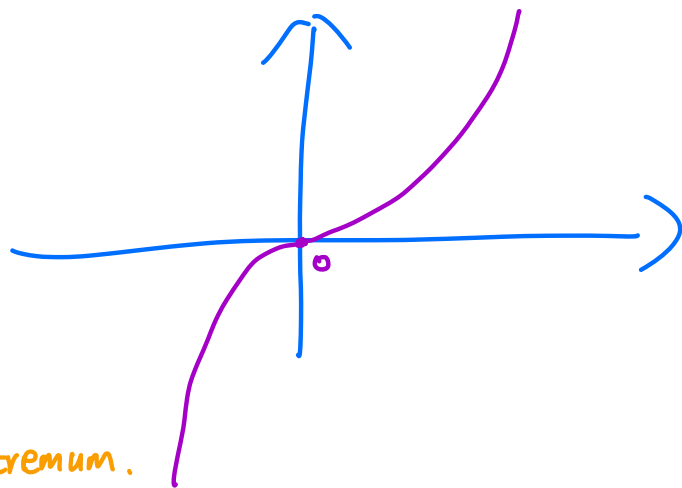
that a local extremum must occur at c .

e.g. $f(x) = x^3$

$$\cdot f'(c) \big|_{c=0} = 3c^2 \big|_{c=0} = 0$$

$\Rightarrow c=0$ is a critical point

$\Rightarrow$ But $c=0$ is NOT the local extremum.



Thms 4.1.1 - 4.1.2 provide a procedure to find absolute extrema of a continuous function on $[a, b]$.

Step 1 . Evaluate f at all critical points and endpoints

$\swarrow$ $\downarrow$

either $f'(c) = 0$; a, b .

or $f'(c)$ DNE finitely

Step 2 : Take the largest and smallest of these values.

$$f(a), f(b), f(c_1), f(c_2), \dots$$

e.g. 4. Find all absolute extrema (with values and positions) of:

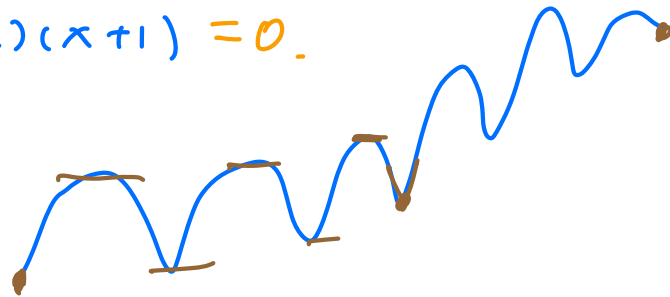
$$f: [-2, 4] \rightarrow \mathbb{R}. \quad f(x) = 2x^3 - 3x^2 - 12x + 15.$$

Sol': Step 1. Solve the critical points!

In the interval $[-2, 4]$, all the points are differentiable, so we only solve the points with $f'(c) = 0$.

Let $f'(x) = 6x^2 - 6x - 12 = 6(x-2)(x+1) = 0$.

$$\therefore C_1 = 2 \text{ or } C_2 = -1.$$



Step 2 : list all the critical points and endpoints
and compare their values.

endpoints : $a = -2$, $b = 4$.

$$\begin{aligned}\text{Hence, } f(-2) &= 2x^3 - 3x^2 - 12x + 15 \Big|_{x=-2} \\ &= -16 - 12 + 24 + 15 = 11\end{aligned}$$

$$\begin{aligned}f(4) &= 2x^3 - 3x^2 - 12x + 15 \Big|_{x=4} \\ &= 128 - 48 - 48 + 15 \\ &= 47 \checkmark \text{ largest}\end{aligned}$$

critical points : ,

$$\begin{aligned}f(2) &= 2x^3 - 3x^2 - 12x + 15 \Big|_{x=2} \\ &= 16 - 12 - 24 + 15 = -5 \text{ smallest}\end{aligned}$$

$$f(-1) = 2x^3 - 3x^2 - 12x + 15 \Big|_{x=-1}$$

$$= -2 - 3 + 12 + 15$$

$$= 22$$

$\therefore$ the absolute max value is 47 at $x = 4$;

min value is -5 at $x = 2$.

□.

